

The Electronics Wall: Why a Self-Replicating Space Factory Cannot Close on Its Own Chips

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Abstract—A self-replicating space factory lands with a fixed cargo and builds copies of itself from local material. Its promise is leverage: the fraction of its own mass it can remake locally, its closure, sets how much installed capacity each launched kilogram becomes, exactly $1/(1 - C)$. We show why that fraction stops short of one at a wall built of chips. The wall is twofold. Chips sit at the end of the deepest supply chain on Earth, and they are staggeringly energy-hungry to make, with an embodied energy roughly three orders of magnitude above smelted metal. Modern seed-factory studies converge on achievable closure near 70 to 96 percent and agree that chasing 100 percent is not worthwhile, so imported electronics are a permanent design feature rather than a temporary compromise. Letting a highly closed seed make its own chips pays off only when it is also swimming in power; starved of power it backfires. Imported electronics, and the launch-mass leverage they cap, are therefore intrinsic to the concept.

Index Terms—self-replication, in-situ manufacturing, material closure, embodied energy, space systems

I. INTRODUCTION

The idea of a self-replicating space factory is old and simple to state: land one factory on the Moon, an asteroid, or Mars, and let it build copies of itself from local material, so that an entire industry grows from a single rocket’s worth of cargo. The canonical study is NASA’s seed-factory design, which framed the concept in terms of “vitamins”: the factory makes almost all of itself locally and imports only the parts it cannot make [1]. The modern synthesis restated the same idea as *closure* - the degree to which a machine can reproduce its own matter, energy, and information - and identified the dynamic that matters most: below full closure the factory waits on resupply, and only near full closure does its growth become self-sustaining [2].

This paper isolates one question inside that framing. If closure is the whole game, what stops it from reaching one? The answer, consistently, is chips. A lone factory can smelt metal, cast structure, and assemble mechanisms, but it cannot make its own electronics. We argue that this is not an incidental gap to be closed later but a wall with two independent foundations - supply-chain depth and embodied energy - and that the wall is exactly what bounds the launch-mass leverage that motivates self-replication in the first place.

II. THE CLOSURE MODEL

We treat the factory as a bill of materials and define its closure C as the mass fraction of its own parts it can manufacture on site. The complement, $1 - C$, is the mass

fraction that must be imported as vitamins for every kilogram the factory builds. Mass balance makes this exact rather than approximate: to add one kilogram of installed capacity, the factory must be supplied with $1 - C$ kilograms it cannot make.

The literature does not place C at one. The canonical seed-factory study estimated that 90-96% mass closure is achievable and argued that chasing the last few percent to full closure is not worthwhile [1]. A recent revival of the concept is more conservative, taking a near-term-realistic closure of about 70% with chips, solar cells, and circuitry staying Earth-sourced [3]. The modern synthesis agrees on the principle: because the marginal part gets disproportionately harder to make as closure rises, the sensible target is high-but-incomplete closure, not one [2]. Across these designs the conclusion is the same. Imported electronics are a permanent design feature, not a temporary compromise on the way to full autonomy. The interesting question is therefore not *whether* a factory imports electronics but *why* that particular part is the one it cannot make, and what the resulting cap on closure costs.

III. THE ELECTRONICS WALL

Chips are the part a lone factory cannot make, and the reason is twofold. The first foundation is supply-chain depth. Semiconductors sit at the end of the deepest supply chain on Earth: more than 400 process steps, materials purified to nine-nines purity, and a global monopoly on the extreme-ultraviolet lithography that patterns the smallest features [4]. No single factory, however well closed on structure, reproduces that pipeline from local rock.

The second foundation is energy, and it is the one that turns a supply-chain argument into a physical wall. The embodied energy of compute logic is on the order of 8,000 kWh/kg, against roughly 5 kWh/kg for smelted metal - a gap of about three orders of magnitude [5], [6]. A factory that would make its own chips must therefore pour vastly more electricity into a kilogram of logic than into a kilogram of the structure that houses it.

These two foundations combine into a single conditional. A factory escapes the electronics wall only if it is both highly self-sufficient *and* swimming in power, and the coupling is sharp. For a highly closed lunar seed at 97% closure, letting it make its own chips locally cuts the time to reach a target scale from about 29 years to about 17 years, but only when it is fed on the order of 4 MW; at about 1 MW, making chips locally backfires, and the factory runs out of electricity before it runs

out of parts [1]. Power is not a free knob. The energy cost of the chips is what decides whether local production helps or hurts, and only a factory already swimming in power comes out ahead.

IV. LAUNCH-MASS LEVERAGE

The reason the closure cap matters economically is leverage. Because mass balance forces $1 - C$ kilograms of vitamins to be imported per kilogram the factory builds, the installed mass per launched mass is exactly $1/(1 - C)$. The function is unforgiving near one: at 67% closure each launched kilogram becomes about $3\times$ its mass in installed capacity, while at 97% closure it becomes about $33\times$ [7]. The last few points of closure buy most of the leverage, which is precisely why the electronics wall - by holding closure below one - caps the payoff of the whole scheme.

The alternative to building in place is shipping a finished installation, and the rocket equation makes that alternative exponentially expensive. The propellant mass ratio grows as $\exp(\Delta v/v_e)$, so the cost of delivering finished mass to a destination rises exponentially with the mission's delta- v [7]. That exponential is the entire economic case for self-replication: it is far cheaper to send a small, high-closure seed and let $1/(1 - C)$ leverage do the rest than to launch the installed capacity directly. The electronics wall does not overturn this case; it bounds it. The imported electronics that hold C below one are exactly the mass that must still be flown, at the rocket equation's exponential price, for every generation the factory builds.

V. DISCUSSION

The picture that emerges is consistent across every part of the argument. Closure drives the economics through $1/(1 - C)$ leverage [7]; closure stops short of one at the electronics wall; and the wall stands on two independent foundations, a supply chain too deep to reproduce locally [4] and an embodied energy some three orders of magnitude above ordinary structure [5], [6]. Because both foundations are physical rather than a matter of engineering maturity, imported electronics are best treated as a permanent line item. Modern seed-factory designs already do so, placing achievable closure between about 70% and 90-96% and declining to chase the rest [1]–[3].

The one escape - having the factory make its own chips - is real but narrow. It helps only for a factory that is both highly closed and richly powered, and it can actively hurt an otherwise capable seed that is power-starved [1]. The practical conclusion is not that self-replication fails, but that its design target is a factory closed on nearly everything except chips, fed by imported vitamins, and sized so that its power budget never tempts it into making the one part it cannot afford to make. The leverage of self-replication is real; the electronics wall is what bounds it, and knowing where the wall stands is what lets a seed be designed to live with it rather than break against it.

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